COHOMOLOGY OF A PROFINITE GROUP IN POSITIVE DEGREES IS TORSION AND VANISHES ON Q-VECTOR SPACES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument for both parts: part (1) is reduced by dimension shifting and then computed, and part (2) is the author's own one-line consequence of part (1).

Editorial note. The label lemma-profinite-group-cohomology-torsion occurs in no other staged Stacks chapter, so no cross-chapter restatement had to be weighed for this card.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- Continuous cohomology of a profinite group and its long exact sequence. Same chapter, the definition of H^i(G, M) for a profinite group G and the associated long exact cohomology sequence. Supplies the functors H^i(G, -) and the exact sequence from which the element of H^1(G, M) is read off.
- Dimension shifting. Same chapter, the dimension shifting argument for the cohomology of a profinite group. Reduces statement (1) in degrees i > 0 to the case i = 1, which is the only degree the counted proof computes.
- Injective objects of the category of G-modules. Same chapter, the existence of enough injectives in the category of G-modules. Supplies the embedding 0 -> M -> I -> N -> 0 with I injective that the proof uses to represent classes in H^1(G, M).
- Open stabilisers in a profinite group. Same chapter, the stabiliser of an element of a discrete G-module under the action of a profinite group is open and hence of finite index. Turns the stabiliser U of the chosen lift x into a finite-index subgroup, producing the integer r that kills the cohomology class.

Lemma 0.1. *Let G be a profinite topological group. Then*

(1) *$H^i(G, M)$ is torsion for $i > 0$ and any G-module M, and*
(2) *$H^i(G, M) = 0$ if M is a $\mathbf{Q}$-vector space.*

Proof. Proof of (1). By dimension shifting we see that it suffices to show that $H^1(G, M)$ is torsion for every G-module M. Choose an exact sequence $0 \to M \to I \to N \to 0$ with I an injective object of the category of G-modules. Then any element of $H^1(G, M)$ is the image of an element $y \in N^G$. Choose $x \in I$ mapping

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to y. The stabilizer $U \subset G$ of x is open, hence has finite index r. Let $g_1, \ldots, g_r \in G$ be a system of representatives for G/U. Then $\sum g_i(x)$ is an invariant element of I which maps to ry. Thus r kills the element of $H^1(G, M)$ we started with. Part (2) follows as then $H^i(G, M)$ is both a $\mathbf{Q}$-vector space and torsion. $\square$